

Security in cloud Networking

Task-1.In AWS, create a new security group named InstaClone-SG and configure it to allow incoming HTTP (port 80) and HTTPS (port 443) traffic only from your own IP address.

Objective

Allow **HTTP (80)** and **HTTPS (443)** access **only from your own IP address**.

Step 1: Open Security Groups

1. Log in to the **AWS Management Console**.
2. Search for **EC2**.
3. Open **EC2**.
4. In the left menu, find **Network & Security → Security Groups**.
5. Click **Create security group**.

Step 2: Configure the Security Group

Enter:

Security group name:

InstaClone-SG

Description:

Security group for InstaClone web server

VPC: Select your required VPC.

Step 3: Add Inbound Rules

Under **Inbound rules**, add:

Rule 1 – HTTP

Field	Value
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Type	HTTP
------	------

Protocol	TCP
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Field	Value
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Port	80
------	----

Source	My IP
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Rule 2 – HTTPS

Field	Value
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Type	HTTPS
------	-------

Protocol	TCP
----------	-----

Port	443
------	-----

Source	My IP
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When you select **My IP**, AWS automatically detects your current public IP and creates a /32 rule.

For example:

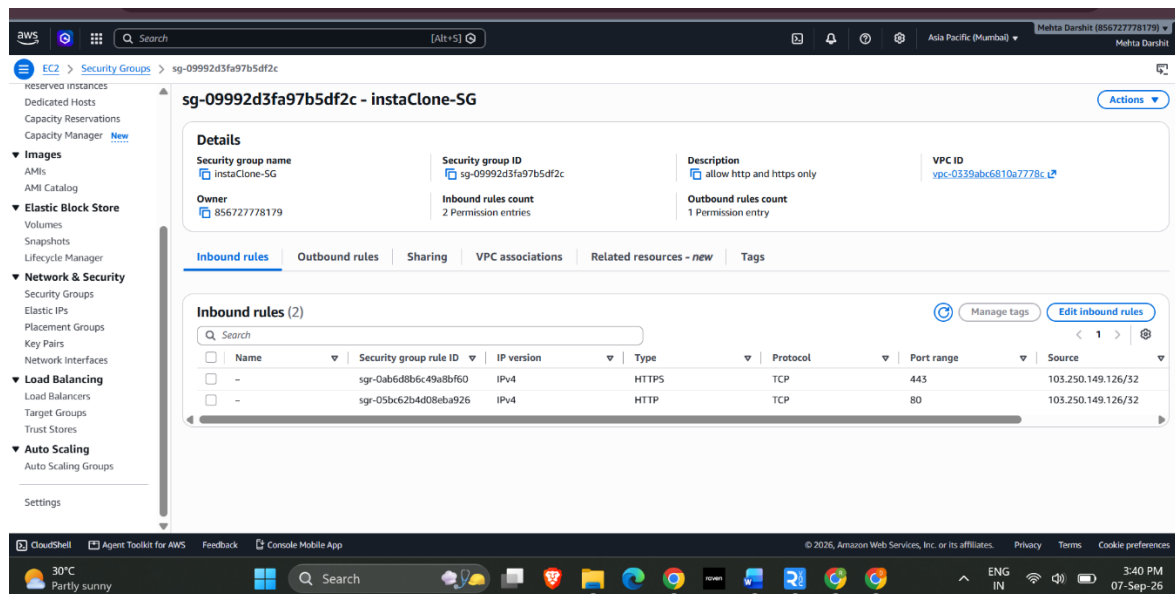
203.0.113.25/32

Step 4: Outbound Rules

For this task, you only need to restrict **incoming** traffic. You can leave the default outbound rule:

All traffic → 0.0.0.0/0

Click **Create security group**.



Task-2. Set up a security group for a mock Zomato-style backend server that allows incoming MySQL (port 3306) connections only from a specific IP address and blocks all other inbound traffic. Use the 'Custom TCP' rule and specify the allowed IP in CIDR format.

Under **Inbound rules**, click **Add rule**.

Select:

Field **Value**

Type Custom TCP

Protocol TCP

Port 3306

Source Custom

CIDR 203.0.113.50/32

The /32 means **only that single IP address**.

For example:

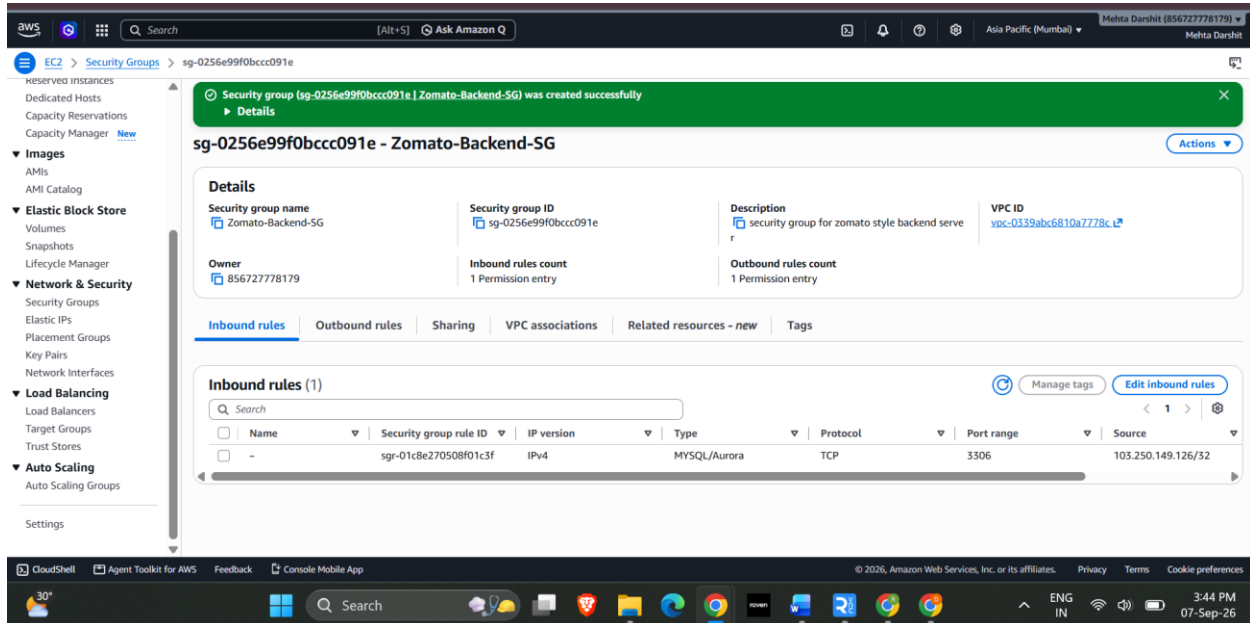
Custom TCP

Port: 3306

Source: 203.0.113.50/32

Do **not** add HTTP, HTTPS, SSH, or other inbound rules for this task.

Click **Create security group**



Task-3. Modify an existing security group to block all outbound traffic except for DNS (port 53) and HTTPS (port 443) to simulate a strict Paytm wallet server environment.

Objective

Allow the server to make outbound connections only for:

- **DNS → Port 53**
- **HTTPS → Port 443**

This simulates a strict Paytm wallet server.

Important: DNS commonly uses **UDP 53**, and TCP 53 can also be used in some circumstances. For a simple lab, add UDP 53 and TCP 443. If you want DNS over TCP as well, add a TCP 53 rule.

Step 1: Select an Existing Security Group

Go to:

EC2 → Security Groups

Select the security group you want to modify.

Go to:

Outbound rules → Edit outbound rules

Step 2: Remove the Existing Rule

You will probably see the default:

All traffic → 0.0.0.0/0

Click **Delete** for this rule.

This is important because otherwise the server can still make unrestricted outbound connections.

Step 3: Add DNS Rule

Click **Add rule**.

Select:

Field	Value
Type	Custom UDP
Protocol	UDP
Port	53
Destination	0.0.0.0/0

Step 4: Add HTTPS Rule

Click **Add rule** again.

Select:

Field	Value
Type	HTTPS
Protocol	TCP

Field Value

Port 443

Destination 0.0.0.0/0

Your outbound rules should now look approximately like:

DNS UDP 53 0.0.0.0/0

HTTPS TCP 443 0.0.0.0/0

Click **Save rules**.

Outbound security group rules successfully modified on security group (sg-09992d3fa97b5df2c | instaClone-SG)

sg-09992d3fa97b5df2c - instaClone-SG

Details

Security group name instaClone-SG	Security group ID sg-09992d3fa97b5df2c	Description allow http and https only	VPC ID vpc-0339abc6810a7778c
Owner 856727778179	Inbound rules count 2 Permission entries	Outbound rules count 2 Permission entries	

Outbound rules (2)

Name	Security group rule ID	IP version	Type	Protocol	Port range	Destination
-	sg-011c351ad00584809	IPv4	DNS (UDP)	UDP	53	0.0.0.0/0
-	sg-0a145dd2cfb563647	IPv4	HTTPS	TCP	443	0.0.0.0/0

Task-4. Compare security groups and network ACLs by listing two key differences in how they control traffic for a Flipkart-like cloud application. Think about the level (instance vs subnet) and direction (inbound/outbound) each operates at.

Feature Security Group

Network ACL

Level Works at the **instance/ENI level**

Works at the **subnet level**

Feature	Security Group	Network ACL
Traffic direction	Controls both inbound and outbound traffic	Controls both inbound and outbound traffic separately
State	Stateful – return traffic is automatically allowed	Stateless – return traffic must be explicitly allowed
Rules	Primarily allow rules; unmatched traffic is implicitly denied	Supports both ALLOW and DENY rules
Typical use	Protect individual EC2 servers	Provide an additional subnet-level security layer

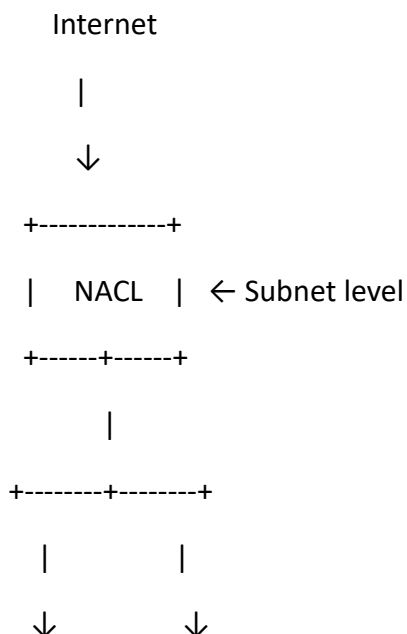
Two key differences

1. Instance vs Subnet

A Security Group is attached to an EC2 instance/network interface and controls traffic to that resource. A Network ACL is associated with a subnet and controls traffic entering or leaving that entire subnet.

2. Stateful vs Stateless

Security Groups are stateful, meaning return traffic for an allowed connection is automatically permitted. Network ACLs are stateless, so inbound and outbound traffic must be allowed separately.



+-----+	+-----+
EC2	EC2
SG	SG ← Instance level
+-----+	+-----+